

APARNA RAJPUT

Toronto, Ontario M2J0A7 | +1 (514) 292-8171
aparna.ar.rajput@gmail.com | linkedin.com/in/aparna-rajput-956a16123

EDUCATION

PhD in Mathematics — Concordia University, Montreal, Canada *(Sept 2019 – Nov 2025)*

GPA: 3.88 / 4.3

Master of Science in Mathematics — Dr. B. R. Ambedkar NIT Jalandhar, Punjab, India *(Aug 2015 – May 2017)*

GPA: 8.16 / 10

Bachelor of Science in Mathematics (Honors) — Janki Devi Memorial College, Delhi University, India *(Aug 2011 – May 2014)*

85.6%

RESEARCH INTERESTS

My research focuses on dynamical systems and operator-theoretic approaches to understanding the long-term behavior of complex systems. In particular, I study ergodic properties of interval maps, quasi-compactness of transfer operators, and the existence of absolutely continuous invariant measures (ACIMs), with an emphasis on how distributions evolve under repeated transformations. I am particularly interested in connecting these ideas to statistical modeling and data-driven approaches.

RESEARCH EXPERIENCE

Ph.D. Researcher | Concordia University

Sept 2019 – Nov 2025

- Research on quasi-compactness of the Frobenius–Perron operator for two classes of interval maps.

Master's Thesis Project | NIT Jalandhar

Aug 2015 – May 2017

- Implemented Maple and Mathematica to solve Legendre's differential equation using the Frobenius method.

Research Intern | IIT Roorkee

May – July 2016

- Research on controllability of delay differential equations using Maple and MATLAB.
 - Developed mathematical modeling programs using Wolfram Mathematica as part of undergraduate coursework and projects.
-

TEACHING EXPERIENCE

Teaching Assistant & Marker | Concordia University

2019 – 2025

- MATH 201 – Calculus
- MATH 201 EC – Elementary Functions
- MATH 203 – Differential & Integral Calculus
- MATH 206 – Algebra & Functions
- MATH 364 – Analysis I
- MATH 365 – Analysis II

- MATH 370 – Differential Equations
- MATH 475 – Discrete Dynamical Systems
- Responsibilities: grading assignments, assisting students with proofs and problem solving, supporting course instruction.

Content Writer | Atharva Academy

- Developed mathematics content for students preparing for college entrance examinations.

PUBLICATIONS

Góra, P., **Rajput, A.** (2026). Quasi-compactness of Frobenius–Perron operator for piecewise convex maps with countably many branches. *Dynamical Systems*.

Rajput, A., Góra, P. (2025). Quasi-compactness of Frobenius–Perron operator for piecewise expanding $C^{1+\varepsilon}$ maps of an interval. *Stochastics and Dynamics*.

AWARDS, HONORS & CERTIFICATIONS

Concordia International Tuition Award of Excellence — Concordia University, 2019–2023
Competitive award covering international tuition fees (value: CAD \$37,915 for eight terms).

Faculty of Arts and Science Fellowship — Concordia University, 2019–2023
Graduate fellowship awarded to support doctoral research (value: \$14,000 per year).

Certificate: Recent Trends in Applied Mathematics — NIT Jalandhar, 2015

Certificate: Advances of Statistics in Science and Engineering — NIT Jalandhar, 2016

Mrs. Manohar Lal Silver Medal — Janki Devi Memorial College, University of Delhi, 2014
Awarded for being the most outstanding student in Mathematics.

Certificate of Merit (Highest Marks in Mathematics Honours) — Janki Devi Memorial College, University of Delhi, 2014

SKILLS

Languages: Python

Software: Wolfram Mathematica, Maple, MATLAB, Maxima